

Linear Mixed Models

Introduction

1.1 Setting

Suppose that N total observations are grouped into K clusters. Let y_{ik} denote the i th outcome in the k th cluster. Group the N_k outcomes in the k th cluster to form the response vector $\mathbf{y}_k = (Y_{k1}, \dots, Y_{kN_k})$. Associate with Y_{ki} a $J \times 1$ vector $\mathbf{x}_{ki}$ of *fixed effect covariates*, and an $L \times 1$ vector $\mathbf{z}_{ki}$ of *random effect covariates*. Structure the covariates into design matrices $\mathbf{X}_k$ and $\mathbf{Z}_k$. Observations Y_{k_1i} and Y_{k_2i} belonging to distinct clusters are independent. However, observations Y_{ki_1} and Y_{ki_2} within a given cluster are potentially dependent.

1.2 Model

Definition 1.1. A linear mixed effect model for $\mathbf{y}_k$ takes the form:

$$\begin{aligned}\mathbf{y}_k &= \mathbf{X}_k\boldsymbol{\beta} + \mathbf{Z}_k\boldsymbol{\gamma}_k + \boldsymbol{\epsilon}_k, \\ \boldsymbol{\gamma}_k &\sim (\mathbf{0}, \mathbf{G}) \perp \boldsymbol{\epsilon}_k \sim (\mathbf{0}, \mathbf{R}_k).\end{aligned}$$

Here $\boldsymbol{\beta}$ is *fixed effect* in the sense that its value is constant across clusters. $\boldsymbol{\gamma}_k$ is a *random effect* whose value varies across clusters according to a distribution with mean zero and covariance $\mathbf{G}$. $\boldsymbol{\epsilon}_k$ is a *residual* whose distribution has mean zero and cluster-specific covariance $\mathbf{R}_k$. ■

Remark 1.1. In contrast to a generalized linear mixed model, an LMM allows for residual covariance between observations Y_{ki} and Y_{kj} :

$$\text{Cov}(Y_{ki}, Y_{kj} | \mathbf{X}_k, \mathbf{Z}_k, \boldsymbol{\gamma}_k) = \mathbf{R}_{k,ij}.$$

◆

1.3 Notation

The components of an LMM are summarized here:

Structures	Dimension	Description
β	$J \times 1$	Fixed effect
γ_k	$L \times 1$	Cluster-specific random effect
ϵ_k	$N_k \times 1$	Residual
α	$M \times 1$	Covariance parameters
$G(\alpha)$	$L \times L$	Random effect covariance
$R_k(\alpha)$	$N_k \times N_k$	Residual covariance

Let $N = \sum_{k=1}^K N_k$. Define the following data structures:

Structure	Dimension
$\mathbf{y} = \text{vec}(\mathbf{y}_1, \dots, \mathbf{y}_K)'$	$N \times 1$
$\mathbf{X} = \text{rbind}(\mathbf{X}_1, \dots, \mathbf{X}_K)$	$N \times J$
$\mathbf{Z} = \text{diag}(\mathbf{Z}_1, \dots, \mathbf{Z}_K)$	$N \times KL$
$\mathcal{G} = \mathbf{I}_{K \times K} \otimes G(\alpha)$	$KL \times KL$
$\mathcal{R} = \text{diag}\{\mathbf{R}_1(\alpha), \dots, \mathbf{R}_K(\alpha)\}$	$N \times N$

In compact notation, the LMM is expressible as:

$$\begin{aligned} \mathbf{y} &= \mathbf{X}\beta + \mathbf{Z}\gamma + \epsilon, \\ \gamma &\sim (\mathbf{0}, \mathcal{G}) \perp \epsilon \sim (\mathbf{0}, \mathcal{R}). \end{aligned} \tag{1.1}$$

Here γ is the $KL \times 1$ vector $\text{vec}(\gamma_1, \dots, \gamma_K)$ and ϵ is the $N \times 1$ vector $\text{vec}(\epsilon_1, \dots, \epsilon_K)$.

1.4 Likelihood

Proposition 1.1. Let $\mathcal{D}_k$ denote the collection covariates relevant to $\mathbf{y}_k$. For any LMM, the likelihood is expressible as:

$$L(\beta, \alpha) = \prod_{k=1}^K \int f(\mathbf{y}_k | \mathcal{D}_k, \gamma_k; \beta, \alpha) f(\gamma_k | \alpha) d\gamma_k$$

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Proof. Factoring the likelihood across clusters:

$$L(\boldsymbol{\beta}, \boldsymbol{\alpha}) = f(\mathbf{y}|\mathcal{D}; \boldsymbol{\beta}, \boldsymbol{\alpha}) = \prod_{k=1}^K f(\mathbf{y}_k|\mathcal{D}_k; \boldsymbol{\beta}, \boldsymbol{\alpha})$$

Introducing the random effect:

$$L(\boldsymbol{\beta}, \boldsymbol{\alpha}) = \prod_{k=1}^K \int f(\mathbf{y}_k, \boldsymbol{\gamma}_k|\mathcal{D}_k; \boldsymbol{\beta}, \boldsymbol{\alpha}) d\boldsymbol{\gamma}_k = \prod_{k=1}^K \int f(\mathbf{y}_k|\mathcal{D}_k, \boldsymbol{\gamma}_k; \boldsymbol{\beta}, \boldsymbol{\alpha}) f(\boldsymbol{\gamma}_k|\boldsymbol{\alpha}) d\boldsymbol{\gamma}_k.$$

■

Marginal Model Approach

Assumption 2.1. Hereafter, independent normal distributions are assumed for the random effects and residuals:

$$\begin{aligned} \mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\epsilon}, \\ \boldsymbol{\gamma} &\sim N(\mathbf{0}, \mathcal{G}) \perp \boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathcal{R}). \end{aligned} \tag{2.1}$$

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Proposition 2.1. The induced marginal model for $\mathbf{y}$ from (1.1) is:

$$\mathbf{y} | (\mathbf{X}, \mathbf{Z}) \sim N(\mathbf{X}\boldsymbol{\beta}, \boldsymbol{\Sigma}), \tag{2.2}$$

where $\boldsymbol{\Sigma} \equiv \mathbb{V}(\mathbf{y}|\mathcal{D}) = \mathcal{R} + \mathbf{Z}\mathcal{G}\mathbf{Z}'$.

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Proof. Since $\boldsymbol{\gamma}$ and $\boldsymbol{\epsilon}$ are normally distributed, the distribution of $\mathbf{y}$ integrated w.r.t. $\boldsymbol{\gamma}$ is again normal. By iterated expectation, the mean is:

$$\mathbb{E}(\mathbf{y}|\mathcal{D}) = \mathbb{E}\{\mathbb{E}(\mathbf{y}|\boldsymbol{\gamma}, \mathcal{D})|\mathcal{D}\} = \mathbb{E}\{\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}\} = \mathbf{X}\boldsymbol{\beta}.$$

By law of total variance, the covariance of $\mathbf{y}$ is:

$$\boldsymbol{\Sigma} \equiv \mathbb{V}(\mathbf{y}|\mathcal{D}) = \mathbb{E}\{\mathbb{V}(\mathbf{y}|\boldsymbol{\gamma}, \mathcal{D})|\mathcal{D}\} + \mathbb{V}\{\mathbb{E}(\mathbf{y}|\boldsymbol{\gamma}, \mathcal{D})|\mathcal{D}\} = \mathcal{R} + \mathbf{Z}\mathcal{G}\mathbf{Z}'.$$

■

Corollary 2.1. The log likelihood of the induced marginal model is:

$$\ell(\boldsymbol{\beta}, \boldsymbol{\alpha}) \propto -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \tag{2.3}$$

♣

2.1 Estimation of β

Result 2.1 (Score for β). The score equation for β is:

$$\mathcal{U}_\beta = \frac{\partial \ell}{\partial \beta} = \mathbf{X}'\Sigma^{-1}(\mathbf{y} - \mathbf{X}\beta).$$

Solving the score equation $\mathcal{U}_\beta \stackrel{\text{Set}}{=} \mathbf{0}$ gives the *generalized least squares* (GLS) estimator:

$$\hat{\beta}(\alpha) = (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{y} = \mathcal{I}_{\beta\beta'}^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{y}. \quad (2.4)$$



Result 2.2 (Information for β). The Hessian of the log likelihood w.r.t. β is:

$$\mathcal{H}_{\beta\beta'} = \frac{\partial^2 \ell}{\partial \beta \partial \beta'} = -\mathbf{X}'\Sigma^{-1}\mathbf{X}$$

The expected information for β is:

$$\mathcal{I}_{\beta\beta'} = \mathbf{X}'\Sigma^{-1}\mathbf{X}. \quad (2.5)$$



2.2 Estimation of α

2.2.1 Profile Likelihood

Remark 2.1. Since the estimator for β is available in closed form, we proceed by forming the profile log likelihood for α , and differentiating to obtain the *efficient score*. 

Definition 2.1. Define the *error projection* $\mathbf{Q}$ as:

$$\mathbf{Q} = \Sigma^{-1} - \Sigma^{-1}\mathbf{X}(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}.$$



Result 2.3 (Error Projection Properties). The error projection has the following properties:

- i. $\mathbf{Q}\mathbf{y} = \Sigma^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta})$.
- ii. $\mathbf{Q}\Sigma\mathbf{Q} = \mathbf{Q}$.



Proof. (i.) Expanding the GLS estimator in the residual $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ gives:

$$\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}} = (\mathbf{I} - \mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}^{-1})\mathbf{y} = \boldsymbol{\Sigma}\mathbf{Q}\mathbf{y}.$$

To establish the second point, expand the right-hand side of $\mathbf{Q}\boldsymbol{\Sigma}\mathbf{Q}$:

$$\mathbf{Q}\boldsymbol{\Sigma}\mathbf{Q} = \mathbf{Q} - \mathbf{Q}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}\boldsymbol{\Sigma}^{-1}.$$

The conclusion holds if the second term vanishes. Expanding the second term using the definition of the error projection gives:

$$\begin{aligned} \mathbf{Q}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}\boldsymbol{\Sigma}^{-1} &= \boldsymbol{\Sigma}^{-1}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}\boldsymbol{\Sigma}^{-1} \\ &\quad - \boldsymbol{\Sigma}^{-1}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}\boldsymbol{\Sigma}^{-1}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}\boldsymbol{\Sigma}^{-1} = \mathbf{0}. \end{aligned}$$

■

Remark 2.2. The following is an identity for differentiation of the error projection $\mathbf{Q}$ w.r.t. a variance component α_p .

$$\frac{\partial \mathbf{Q}}{\partial \alpha_p} = -\mathbf{Q} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \mathbf{Q}.$$

◆

Proposition 2.2. The profile log likelihood for $\boldsymbol{\alpha}$ is:

$$\ell_p(\boldsymbol{\alpha}) \propto -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} \mathbf{y}' \mathbf{Q} \mathbf{y}. \quad (2.6)$$

◆

Proof. The marginal log likelihood for $\mathbf{y}$ is:

$$\ell(\boldsymbol{\beta}, \boldsymbol{\alpha}) \propto -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})' \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

Substituting the GLS estimator (2.4) for $\boldsymbol{\beta}$ into the marginal log likelihood:

$$\ell_p(\boldsymbol{\alpha}) \propto -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})' \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}).$$

Expressing profile log likelihood in terms of the error projection:

$$\begin{aligned} \ell_p(\boldsymbol{\alpha}) &\propto -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} \{ \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \}' \boldsymbol{\Sigma} \{ \boldsymbol{\Sigma}^{-1} (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \} \\ &= -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} \mathbf{y}' \mathbf{Q} \boldsymbol{\Sigma} \mathbf{Q} \mathbf{y} \\ &= -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} \mathbf{y}' \mathbf{Q} \mathbf{y}. \end{aligned}$$

■

2.2.2 Restricted Likelihood

Definition 2.2. A **restricted likelihood** is formed by applying a Jeffreys prior to the fixed effects:

$$\pi(\boldsymbol{\beta}) \propto \det(\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'})^{-1/2}.$$

The restricted log likelihood for $\boldsymbol{\alpha}$ is:

$$\begin{aligned} \ell_r(\boldsymbol{\alpha}) &\equiv \ell_p(\boldsymbol{\alpha}) + \ln \pi(\boldsymbol{\beta}) = \ell_p(\boldsymbol{\alpha}) - \frac{1}{2} \ln \det(\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'}) \\ &\propto -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} \mathbf{y}' \mathbf{Q} \mathbf{y} - \frac{1}{2} \ln \det(\mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X}). \end{aligned} \quad (2.7)$$

■

Remark 2.3. Maximum likelihood estimates (MLEs) of the variance components $\boldsymbol{\alpha}$ are downward biased, whereas the restricted MLEs (ReMLs) are unbiased. ♦

Remark 2.4. The following are identities for differentiation of a matrix $\boldsymbol{\Sigma}(\boldsymbol{\alpha})$ w.r.t. a variance component α_p :

- Derivative of inverse:

$$\frac{\partial}{\partial \alpha_p} \boldsymbol{\Sigma}^{-1} = -\boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \boldsymbol{\Sigma}^{-1}.$$

- Derivative of log determinant:

$$\frac{\partial}{\partial \alpha_p} \ln \det(\boldsymbol{\Sigma}) = \text{tr} \left(\boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \right).$$

♦

Result 2.4 (Restricted Score for $\boldsymbol{\alpha}$). The restricted score for α_p is:

$$\mathcal{U}_{\alpha_p} = \frac{\partial \ell_r}{\partial \alpha_p} = -\frac{1}{2} \text{tr} \left(\mathbf{Q} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \right) + \frac{1}{2} \mathbf{y}' \mathbf{Q} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \mathbf{Q} \mathbf{y}. \quad (2.8)$$

♣

Proof. The derivative of the restricted log likelihood (2.7) w.r.t. α_p is:

$$\begin{aligned} \frac{\partial \ell_r}{\partial \alpha_p} &= -\frac{1}{2} \text{tr} \left(\boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \right) + \frac{1}{2} \mathbf{y}' \mathbf{Q} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \mathbf{Q} \mathbf{y} \\ &\quad + \frac{1}{2} \text{tr} \left((\mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}' \boldsymbol{\Sigma}^{-1} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \boldsymbol{\Sigma}^{-1} \mathbf{X} \right). \end{aligned}$$

Combining the trace terms gives:

$$\frac{\partial \ell_r}{\partial \alpha_p} = -\frac{1}{2} \text{tr} \left(\left\{ \boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1} \mathbf{X} (\mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X})^{-1} \mathbf{X}' \boldsymbol{\Sigma}^{-1} \right\} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \right) + \frac{1}{2} \mathbf{y}' \mathbf{Q} \frac{\partial \boldsymbol{\Sigma}}{\partial \alpha_p} \mathbf{Q} \mathbf{y}.$$

■

Proposition 2.3. $Q\mathbf{y}$ is distributed as:

$$Q\mathbf{y}|\mathcal{D} \sim N(\mathbf{0}, Q).$$

◆

Proof. Since $\mathbf{y}|\mathcal{D} \sim N(\mathbf{X}\boldsymbol{\beta}, \Sigma)$ is normally distributed, the linear function $Q\mathbf{y}$ is again normally distributed with mean:

$$\begin{aligned}\mathbb{E}(Q\mathbf{y}|\mathcal{D}) &= Q\mathbf{X}\boldsymbol{\beta} = \{\Sigma^{-1} - \Sigma^{-1}\mathbf{X}(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\}\mathbf{X}\boldsymbol{\beta} \\ &= \Sigma^{-1}\mathbf{X}\boldsymbol{\beta} - \Sigma^{-1}\mathbf{X}\boldsymbol{\beta} = \mathbf{0},\end{aligned}$$

and variance:

$$\mathbb{V}(Q\mathbf{y}|\mathbf{X}) = Q\mathbb{V}(\mathbf{y}|\mathbf{X})Q = Q\Sigma Q = Q.$$

■

Proposition 2.4. Suppose $\mathbb{E}(\mathbf{y}) = \boldsymbol{\mu}$ and $\mathbb{V}(\mathbf{y}) = \Sigma$. The expectation of the quadratic form $\mathbf{y}'\mathbf{A}\mathbf{y}$ is:

$$\mathbb{E}(\mathbf{y}'\mathbf{A}\mathbf{y}) = \text{tr}(\Sigma\mathbf{A}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}.$$

◆

Proof.

$$\begin{aligned}\mathbb{E}(\mathbf{y}'\mathbf{A}\mathbf{y}) &= \mathbb{E}\{\text{tr}(\mathbf{y}'\mathbf{A}\mathbf{y})\} = \mathbb{E}\{\text{tr}(\mathbf{y}\mathbf{y}'\mathbf{A})\} = \text{tr}\{\mathbb{E}(\mathbf{y}\mathbf{y}')\mathbf{A}\} \\ &= \text{tr}\{(\Sigma + \boldsymbol{\mu}\boldsymbol{\mu}')\mathbf{A}\} = \text{tr}(\Sigma\mathbf{A}) + \text{tr}(\boldsymbol{\mu}\boldsymbol{\mu}'\mathbf{A}) = \text{tr}(\Sigma\mathbf{A}) + \boldsymbol{\mu}'\mathbf{A}\boldsymbol{\mu}.\end{aligned}$$

■

Result 2.5 (Restricted Information for $\boldsymbol{\alpha}$). The restricted information between α_p and α_q is:

$$\mathcal{I}_{\alpha_p\alpha_q} = \frac{1}{2}\text{tr}\left(Q\frac{\partial\Sigma}{\partial\alpha_p}Q\frac{\partial\Sigma}{\partial\alpha_q}\right). \quad (2.9)$$

♣

Proof. Differentiating the restricted score (2.8) to obtain the Hessian:

$$\begin{aligned}\mathcal{H}_{\alpha_p\alpha_q} &\equiv \frac{\partial^2\ell_r}{\partial\alpha_p\partial\alpha_q} \\ &= +\frac{1}{2}\text{tr}\left(Q\frac{\partial\Sigma}{\partial\alpha_q}Q\frac{\partial\Sigma}{\partial\alpha_p}\right) - \frac{1}{2}\text{tr}\left(Q\frac{\partial^2\Sigma}{\partial\alpha_q\partial\alpha_p}\right) \\ &\quad - \frac{1}{2}\mathbf{y}'Q\frac{\partial\Sigma}{\partial\alpha_q}Q\frac{\partial\Sigma}{\partial\alpha_p}Q\mathbf{y} + \frac{1}{2}\mathbf{y}'Q\frac{\partial^2\Sigma}{\partial\alpha_q\partial\alpha_p}Q\mathbf{y} - \frac{1}{2}\mathbf{y}'Q\frac{\partial\Sigma}{\partial\alpha_p}Q\frac{\partial\Sigma}{\partial\alpha_q}Q\mathbf{y}.\end{aligned}$$

Taking the expectation:

$$\begin{aligned}\mathbb{E}(\mathcal{H}_{\alpha_p\alpha_q}|\mathbf{X}) &= \frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_q}\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_p}\right) - \frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial^2\boldsymbol{\Sigma}}{\partial\alpha_q\partial\alpha_p}\right) \\ &\quad - \frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_q}\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_p}\right) + \frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial^2\boldsymbol{\Sigma}}{\partial\alpha_p\partial\alpha_q}\right) - \frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_p}\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_q}\right).\end{aligned}$$

Combining like terms gives the result. ■

Definition 2.3. Suppose the covariance matrix $\boldsymbol{\Sigma}$ is linear in the parameters $\boldsymbol{\alpha}$ s.t.

$$\frac{\partial^2\boldsymbol{\Sigma}}{\partial\alpha_p\partial\alpha_q} = \mathbf{0}.$$

In this setting, the observed information is:

$$\mathcal{J}_{\alpha_p\alpha_q} = -\frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_q}\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_p}\right) + \mathbf{y}'\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_p}\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_q}\mathbf{Q}\mathbf{y}.$$

The **average information** is defined as:

$$\mathcal{A}_{\alpha_p\alpha_q} = \frac{1}{2}(\mathcal{I}_{\alpha_p\alpha_q} + \mathcal{J}_{\alpha_p\alpha_q}) = \frac{1}{2}\mathbf{y}'\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_p}\mathbf{Q}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_q}\mathbf{Q}\mathbf{y}. \quad (2.10)$$
■

Remark 2.5. The *Newton-Raphson* iteration for estimation of the variance components is:

$$\boldsymbol{\alpha}^{(r+1)} = \boldsymbol{\alpha}^{(r)} + \mathcal{A}^{-1}(\boldsymbol{\alpha}^{(r)})\mathcal{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}^{(r)}),$$

where $\boldsymbol{\alpha}^{(r)}$ is the current parameter estimate, $\mathcal{A}$ is the average information for $\boldsymbol{\alpha}$ from (2.10), and $\mathcal{U}_{\boldsymbol{\alpha}}$ is the restricted score for $\boldsymbol{\alpha}$ from (2.8). ◆

Conditional Model Approach

3.1 Mixed Model Equations

Remark 3.1. In the marginal model approach, $\boldsymbol{\gamma}$ was treated as unobserved data and integrated away. In the conditional model approach, $\boldsymbol{\gamma}$ is treated as a parameter that requires estimation. ◆

Proposition 3.1. The conditional model log likelihood is:

$$\begin{aligned}\ell_C(\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\gamma}) &= \ln f(\mathbf{y}|\mathcal{D}; \boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\gamma}) + \ln f(\boldsymbol{\gamma}; \boldsymbol{\alpha}) \\ &\quad \boldsymbol{\alpha} - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\boldsymbol{\gamma})'\mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\boldsymbol{\gamma}) - \frac{1}{2}\boldsymbol{\gamma}'\mathcal{G}^{-1}\boldsymbol{\gamma}.\end{aligned}$$
◆

Result 3.1 (Mixed Model Equations). For fixed α , the best linear unbiased estimator $\hat{\beta}$ of β , and the best linear unbiased predictor $\hat{\gamma}$ of γ satisfy:

$$\begin{pmatrix} \mathbf{X}'\mathcal{R}^{-1}\mathbf{X} & \mathbf{X}'\mathcal{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathcal{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z} + \mathcal{G}^{-1} \end{pmatrix} \begin{pmatrix} \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathcal{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathcal{R}^{-1}\mathbf{y} \end{pmatrix}. \quad (3.1)$$

♣

Proof. The score equations for β and γ are:

$$\begin{aligned} \mathcal{U}_\beta &= \mathbf{X}'\mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta - \mathbf{Z}\gamma) \stackrel{\text{Set}}{=} 0, \\ \mathcal{U}_\gamma &= \mathbf{Z}'\mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta - \mathbf{Z}\gamma) + \mathcal{G}^{-1}\gamma \stackrel{\text{Set}}{=} 0. \end{aligned}$$

Re-arranging:

$$\begin{aligned} \mathbf{X}'\mathcal{R}^{-1}\mathbf{X}\beta + \mathbf{X}'\mathcal{R}^{-1}\mathbf{Z}\gamma &= \mathbf{X}'\mathcal{R}^{-1}\mathbf{y}, \\ \mathbf{Z}'\mathcal{R}^{-1}\mathbf{X}\beta + (\mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z} + \mathcal{G}^{-1})\gamma &= \mathbf{Z}'\mathcal{R}^{-1}\mathbf{y}. \end{aligned}$$

In matrix format:

$$\begin{pmatrix} \mathbf{X}'\mathcal{R}^{-1}\mathbf{X} & \mathbf{X}'\mathcal{R}^{-1}\mathbf{Z} \\ \mathbf{Z}'\mathcal{R}^{-1}\mathbf{X} & \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z} + \mathcal{G}^{-1} \end{pmatrix} \begin{pmatrix} \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathcal{R}^{-1}\mathbf{y} \\ \mathbf{Z}'\mathcal{R}^{-1}\mathbf{y} \end{pmatrix}.$$

Define $\mathbf{V}$ and $\mathbf{W}$ as:

$$\mathbf{V} = (\mathbf{X}, \mathbf{Z}), \quad \mathbf{W} = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathcal{G} \end{pmatrix}.$$

Now the normal equations are expressible as:

$$(\mathbf{V}'\mathcal{R}^{-1}\mathbf{V} + \mathbf{W}^{-1}) \begin{pmatrix} \beta \\ \gamma \end{pmatrix} = \mathbf{V}'\mathcal{R}^{-1}\mathbf{y}.$$

Hence, the best linear estimates of β, γ are given by:

$$\begin{pmatrix} \hat{\beta} \\ \hat{\gamma} \end{pmatrix} = (\mathbf{V}'\mathcal{R}^{-1}\mathbf{V} + \mathbf{W}^{-1})^{-1} \mathbf{V}'\mathcal{R}^{-1}\mathbf{y}.$$

■

3.2 Random Effect Prediction

Result 3.2 (Empirical Bayes Estimation). The best linear unbiased prediction of γ is given by:

$$\tilde{\gamma} = \mathbb{E}(\gamma|\mathbf{y}, \mathcal{D}) = \mathcal{G}\mathbf{Z}'\Sigma^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathcal{G}\mathbf{Z}'\mathbf{Q}\mathbf{y}. \quad (3.2)$$

♣

Proof. From model (2.1), γ and ϵ are jointly distributed as:

$$\begin{pmatrix} \gamma \\ \epsilon \end{pmatrix} \sim N \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathcal{G} & \mathbf{0} \\ \mathbf{0} & \mathcal{R} \end{pmatrix}.$$

The joint distribution of $\mathbf{y}$ and γ is a linear transformation of $\text{vec}(\gamma, \epsilon)$:

$$\begin{pmatrix} \mathbf{y} \\ \gamma \end{pmatrix} = \begin{pmatrix} \mathbf{X}\beta \\ \mathbf{0} \end{pmatrix} + \begin{pmatrix} \mathbf{Z} & \mathbf{I} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \gamma \\ \epsilon \end{pmatrix}.$$

Thus $\text{vec}(\mathbf{y}, \gamma)$ is normally distributed with mean $\text{vec}(\mathbf{X}\beta, \mathbf{0})$ and variance:

$$\begin{aligned} \mathbb{V} \begin{pmatrix} \mathbf{y} \\ \gamma \end{pmatrix} &= \begin{pmatrix} \mathbf{Z} & \mathbf{I} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathcal{G} & \mathbf{0} \\ \mathbf{0} & \mathcal{R} \end{pmatrix} \begin{pmatrix} \mathbf{Z}' & \mathbf{I} \\ \mathbf{I} & \mathbf{0} \end{pmatrix} \\ &= \begin{pmatrix} \mathbf{Z}\mathcal{G}\mathbf{Z}' + \mathcal{R} & \mathbf{Z}\mathcal{G} \\ \mathcal{G}\mathbf{Z}' & \mathcal{G} \end{pmatrix} \equiv \begin{pmatrix} \Sigma & \mathbf{Z}\mathcal{G} \\ \mathcal{G}\mathbf{Z}' & \mathcal{G} \end{pmatrix}. \end{aligned}$$

The conditional distribution of γ given $\mathbf{y}$ is again normal with expectation and variance:

$$\mathbb{E}(\gamma|\mathbf{y}, \mathcal{D}) = \mathcal{G}\mathbf{Z}'\Sigma^{-1}(\mathbf{y} - \mathbf{X}\beta), \quad \mathbb{V}(\gamma|\mathbf{y}, \mathcal{D}) = \mathcal{G} - \mathcal{G}\mathbf{Z}'\Sigma^{-1}\mathbf{Z}\mathcal{G}.$$

From the Gauss-Markov theorem, the best linear unbiased estimator of β is the generalized least squares estimator:

$$\hat{\beta}(\alpha) = (\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{y}.$$

Substituting $\hat{\beta}(\alpha)$ into $\mathbb{E}(\gamma|\mathbf{y})$ gives:

$$\tilde{\gamma} = \mathcal{G}\mathbf{Z}'\Sigma^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta}) = \mathcal{G}\mathbf{Z}'\mathbf{Q}\mathbf{y}.$$

■

Corollary 3.1. The EB prediction $\tilde{\gamma}$ is a weighted average between the GLS estimator $\hat{\gamma}$ of γ and zero:

$$\tilde{\gamma} = (\mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})^{-1}\{(\mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})\hat{\gamma} + \mathcal{G}^{-1}\mathbf{0}\}.$$

That is, $\tilde{\gamma}$ is a *shrinkage estimator*.

♣

Proof. From the induced marginal model (2.2), $\Sigma = \mathcal{R} + \mathbf{Z}\mathcal{G}\mathbf{Z}'$. Multiplying by $\mathbf{Z}'\mathcal{R}^{-1}$ on the left and rearranging gives:

$$\begin{aligned} \mathbf{Z}'\mathcal{R}^{-1}\Sigma &= \mathbf{Z}'\mathcal{R}^{-1}(\mathcal{R} + \mathbf{Z}\mathcal{G}\mathbf{Z}') = \mathbf{Z}' + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z}\mathcal{G}\mathbf{Z}' = (\mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})\mathcal{G}\mathbf{Z}', \\ (\mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})^{-1}\mathbf{Z}'\mathcal{R}^{-1} &= \mathcal{G}\mathbf{Z}'\Sigma^{-1}. \end{aligned}$$

Suppose γ were treated as a fixed effect and estimated from the model:

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{Z}\gamma + \epsilon,$$

where $\epsilon \sim N(\mathbf{0}, \mathcal{R})$. The BLUE of γ is:

$$\hat{\gamma} = (\mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})^{-1}\mathbf{Z}'\mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta).$$

From (3.2), the EB estimator of γ is:

$$\tilde{\gamma} = \mathcal{G}\mathbf{Z}'\Sigma^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta}).$$

Using the equivalence $\mathcal{G}\mathbf{Z}'\Sigma^{-1} = (\mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})^{-1}\mathbf{Z}'\mathcal{R}^{-1}$, the EB estimator of γ is expressible as:

$$\begin{aligned}\tilde{\gamma} &= (\mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})^{-1}\mathbf{Z}'\mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\hat{\beta}) \\ &= (\mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})^{-1}\{(\mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})\hat{\gamma} + \mathcal{G}^{-1}\mathbf{0}\}.\end{aligned}$$

■

Corollary 3.2. The empirical Bayes prediction of $\mathbf{y}$ is:

$$\tilde{\mathbf{y}} = \mathbf{Z}\mathcal{G}\mathbf{Z}'\Sigma^{-1}\mathbf{y} + (\mathbf{I} - \mathbf{Z}\mathcal{G}\mathbf{Z}'\Sigma^{-1})\mathbf{X}\hat{\beta}.$$

The EB prediction $\tilde{\mathbf{y}}$ is interpretable as a weighted average of the observations $\mathbf{y}$ and the fitted values $\mathbf{X}\hat{\beta}$. ♣

Proof. The first term is expressible as:

$$\mathbf{X}\hat{\beta}(\alpha) = \mathbf{X}(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\mathbf{y} = \Sigma\{\Sigma^{-1}\mathbf{X}(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\}\mathbf{y}.$$

The second term is expressible as:

$$\mathbf{Z}\hat{\gamma} = \mathbf{Z}\mathcal{G}\mathbf{Z}'\mathbf{Q}\mathbf{y} = \mathbf{Z}\mathcal{G}\mathbf{Z}'\{\mathbf{I} - \Sigma^{-1}\mathbf{X}(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}\}\mathbf{y}$$

Combining like terms gives the result. ■

EM Algorithm

Proof. Regarding γ as missing data, the *complete data* log likelihood is:

$$\begin{aligned}\ell(\beta, \alpha) &\propto -\frac{1}{2}\ln\det(\Sigma) - \frac{1}{2}(\mathbf{y} - \mathbf{X}\beta - \mathbf{Z}\gamma)' \mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta - \mathbf{Z}\gamma) - \frac{1}{2}\gamma' \mathcal{G}^{-1}\gamma \\ &= -\frac{1}{2}\ln\det(\Sigma) - \frac{1}{2}(\mathbf{y} - \mathbf{X}\beta)' \mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta) \\ &\quad - \frac{1}{2}\gamma' \mathbf{Z}' \mathcal{R}^{-1} \mathbf{Z} \gamma + 2 \cdot \frac{1}{2} \gamma' \mathbf{Z}' \mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta) - \frac{1}{2}\gamma' \mathcal{G}^{-1}\gamma \\ &= -\frac{1}{2}\ln\det(\Sigma) - \frac{1}{2}(\mathbf{y} - \mathbf{X}\beta)' \mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta) \\ &\quad + \gamma' \mathbf{Z}' \mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\beta) - \frac{1}{2}\gamma'(\mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z})\gamma.\end{aligned}$$

From the derivation of the EB estimator for $\boldsymbol{\gamma}$, the conditional distribution of $\boldsymbol{\gamma}$ given the observed data is normal with mean and covariance:

$$\mathbb{E}(\boldsymbol{\gamma}|\mathbf{y}, \mathcal{D}) = \mathcal{G}\mathbf{Z}'\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}), \quad \mathbb{V}(\boldsymbol{\gamma}|\mathbf{y}, \mathcal{D}) = \mathcal{G} - \mathcal{G}\mathbf{Z}'\boldsymbol{\Sigma}^{-1}\mathbf{Z}\mathcal{G}.$$

The EM objective function is defined as the expectation of the complete data log likelihood given the observed data and the current parameter state:

$$Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) \equiv E\{\ell(\boldsymbol{\beta}, \boldsymbol{\alpha})|\mathbf{y}, \mathcal{D}, \boldsymbol{\theta}^{(r)}\}.$$

Define the following working expectations:

$$\begin{aligned} \hat{\boldsymbol{\gamma}}^{(r)} &\equiv \mathbb{E}(\boldsymbol{\gamma}|\mathbf{y}, \mathcal{D}, \boldsymbol{\theta}^{(r)}) = \mathcal{G}^{(r)}\mathbf{Z}'\boldsymbol{\Sigma}^{(r),-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}^{(r)}), \\ \hat{\mathbf{V}}^{(r)} &\equiv \mathbb{V}(\boldsymbol{\gamma}|\mathbf{y}, \mathcal{D}, \boldsymbol{\theta}^{(r)}) = \mathcal{G}^{(r)} - \mathcal{G}^{(r)}\mathbf{Z}'\boldsymbol{\Sigma}^{(r),-1}\mathbf{Z}\mathcal{G}^{(r)}. \end{aligned}$$

Let $\mathbf{A} \equiv \mathcal{G}^{-1} + \mathbf{Z}'\mathcal{R}^{-1}\mathbf{Z}$. Using the working expectations:

$$\begin{aligned} \mathbb{E}(\boldsymbol{\gamma}'\mathbf{Z}'\mathcal{R}^{-1}\mathbf{y}|\mathbf{y}, \mathcal{D}) &= (\hat{\boldsymbol{\gamma}}^{(r)})'\mathbf{Z}\mathcal{R}^{-1}\mathbf{y}, \\ \mathbb{E}(\boldsymbol{\gamma}'\mathbf{A}\boldsymbol{\gamma}|\mathbf{y}, \mathcal{D}) &= \text{tr}\{\hat{\mathbf{V}}^{(r)}\mathbf{A}\} + (\hat{\boldsymbol{\gamma}}^{(r)})'\mathbf{A}(\hat{\boldsymbol{\gamma}}^{(r)}). \end{aligned}$$

The EM objective function is now expressible as:

$$\begin{aligned} Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(r)}) &= -\frac{1}{2}\ln \det(\boldsymbol{\Sigma}) - \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'\mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \\ &\quad + (\hat{\boldsymbol{\gamma}}^{(r)})'\mathbf{Z}\mathcal{R}^{-1}\mathbf{y} - \frac{1}{2}\text{tr}\{\hat{\mathbf{V}}^{(r)}\mathbf{A}\} - \frac{1}{2}(\hat{\boldsymbol{\gamma}}^{(r)})'\mathbf{A}(\hat{\boldsymbol{\gamma}}^{(r)}). \end{aligned}$$

Consider a conditional maximization approach. Recall that the MLE of $\boldsymbol{\beta}$ is available in closed form. Let $\boldsymbol{\beta}^{(r+1)} \leftarrow \hat{\boldsymbol{\beta}}(\boldsymbol{\alpha}^{(r)})$ denote the GLS estimate of $\boldsymbol{\beta}$ given the current variance components $\boldsymbol{\alpha}^{(r)}$. Score equations for the variance components are obtained by differentiating the EM objective function:

$$\begin{aligned} \mathcal{U}_{\alpha_p}(\boldsymbol{\alpha}|\boldsymbol{\beta}^{(r+1)}, \boldsymbol{\theta}^{(r)}) &= -\frac{1}{2}\text{tr}\left(\boldsymbol{\Sigma}^{-1}\frac{\partial\boldsymbol{\Sigma}}{\partial\alpha_p}\right) + \frac{1}{2}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}^{(r+1)})'\mathcal{R}^{-1}\frac{\partial\mathcal{R}}{\partial\alpha_p}\mathcal{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}^{(r+1)}) \\ &\quad - (\hat{\boldsymbol{\gamma}}^{(r)})'\mathbf{Z}\mathcal{R}^{-1}\frac{\partial\mathcal{R}}{\partial\alpha_p}\mathcal{R}^{-1}\mathbf{y} - \frac{1}{2}\text{tr}\left(\hat{\mathbf{V}}^{(r)}\frac{\partial\mathbf{A}}{\partial\alpha_p}\right) - \frac{1}{2}(\hat{\boldsymbol{\gamma}}^{(r)})'\frac{\partial\mathbf{A}}{\partial\alpha_p}(\hat{\boldsymbol{\gamma}}^{(r)}), \end{aligned}$$

where:

$$\frac{\partial\mathbf{A}}{\partial\alpha_p} = -\mathcal{G}^{-1}\frac{\partial\mathcal{G}}{\partial\alpha_p}\mathcal{G}^{-1} - \mathbf{Z}'\mathcal{R}^{-1}\frac{\partial\mathcal{R}}{\partial\alpha_p}\mathcal{R}^{-1}\mathbf{Z}.$$

The score equations for the variance components are solved numerically to obtain $\boldsymbol{\alpha}^{(r+1)}$. The algorithm iterates between updating $\boldsymbol{\beta}^{(r)}$ and updating $\boldsymbol{\alpha}^{(r)}$ until the improvement $\ell(\boldsymbol{\beta}^{(r+1)}, \boldsymbol{\alpha}^{(r+1)}) - \ell(\boldsymbol{\beta}^{(r)}, \boldsymbol{\alpha}^{(r)})$ in the marginal log likelihood (2.3) falls below the tolerance. ■

Inference

5.1 Fixed Effects

Remark 5.1. Throughout, consider the marginalized LMM:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$. Partition the regression parameter $\boldsymbol{\beta} = (\boldsymbol{\beta}_A, \boldsymbol{\beta}_B)$. Suppose that $H_0 : \boldsymbol{\beta}_A = \boldsymbol{\beta}_A^*$ is of interest. Let $\mathbf{X} = (\mathbf{X}_A, \mathbf{X}_B)$ denote the corresponding partition of the fixed effect design matrix. ◆

Definition 5.1. From (2.5), the information for $\boldsymbol{\beta}$ is $\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'} = \mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X}$. Partition the information as:

$$\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'} = \begin{pmatrix} \mathcal{I}_{\boldsymbol{\beta}_A\boldsymbol{\beta}_A'} & \mathcal{I}_{\boldsymbol{\beta}_A\boldsymbol{\beta}_B'} \\ \mathcal{I}_{\boldsymbol{\beta}_B\boldsymbol{\beta}_A'} & \mathcal{I}_{\boldsymbol{\beta}_B\boldsymbol{\beta}_B'} \end{pmatrix}.$$

The **efficient information** for $\boldsymbol{\beta}_A$ is:

$$\mathcal{I}_{\boldsymbol{\beta}_A\boldsymbol{\beta}_A'|\boldsymbol{\beta}_B} = \mathcal{I}_{\boldsymbol{\beta}_A\boldsymbol{\beta}_A'} - \mathcal{I}_{\boldsymbol{\beta}_A\boldsymbol{\beta}_B'}\mathcal{I}_{\boldsymbol{\beta}_B\boldsymbol{\beta}_B'}^{-1}\mathcal{I}_{\boldsymbol{\beta}_B\boldsymbol{\beta}_A'}.$$

■

Proposition 5.1. The Wald test of $H_0 : \boldsymbol{\beta}_A = \boldsymbol{\beta}_A^*$ is:

$$T_W = (\hat{\boldsymbol{\beta}}_A - \boldsymbol{\beta}_A^*)'\mathcal{I}_{\boldsymbol{\beta}_A\boldsymbol{\beta}_A'|\boldsymbol{\beta}_B}(\hat{\boldsymbol{\beta}}_A - \boldsymbol{\beta}_A^*) \sim \chi_{\dim(\boldsymbol{\beta}_A^*)}^2.$$

◆

Proposition 5.2. Let $\tilde{\boldsymbol{\beta}}_B$ denote a consistent estimate of $\boldsymbol{\beta}_B$ under $H_0 : \boldsymbol{\beta}_A = \boldsymbol{\beta}_A^*$, such as a solution to the score equation:

$$\mathcal{U}_{\boldsymbol{\beta}_B}(\boldsymbol{\beta}_A = \boldsymbol{\beta}_A^*, \boldsymbol{\beta}_B) \stackrel{\text{Set}}{=} \mathbf{0}.$$

Let $\tilde{\mathcal{U}}_{\boldsymbol{\beta}_A}$ denote $\mathcal{U}_{\boldsymbol{\beta}_A}(\boldsymbol{\beta}_A = \boldsymbol{\beta}_A^*, \boldsymbol{\beta}_B = \tilde{\boldsymbol{\beta}}_B)$. The score test of $H_0 : \boldsymbol{\beta}_A = \boldsymbol{\beta}_A^*$ is:

$$T_S = \tilde{\mathcal{U}}_{\boldsymbol{\beta}_A}'\mathcal{I}_{\boldsymbol{\beta}_A\boldsymbol{\beta}_A'|\boldsymbol{\beta}_B}^{-1}\tilde{\mathcal{U}}_{\boldsymbol{\beta}_A} \sim \chi_{\dim(\boldsymbol{\beta}_A^*)}^2.$$

◆

5.2 Variance Components

Example 5.1 (Kernel Regression). Consider the kernel regression model:

$$y_i = \mathbf{x}_i' \boldsymbol{\beta} + f(\mathbf{z}_i) + \epsilon_i,$$

where $\epsilon_i \sim N(0, \sigma^2)$, and $f(\cdot)$ is an unknown function belonging to a reproducing kernel Hilbert space $\mathcal{H}$, with reproducing kernel $k(\cdot, \cdot)$. This model is isomorphic to the following random intercept model:

$$\begin{aligned} \mathbf{y} &= \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\gamma} + \boldsymbol{\epsilon}, \\ \boldsymbol{\gamma} &\sim N(\mathbf{0}, \tau^2 \mathbf{K}) \perp \boldsymbol{\epsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}), \end{aligned}$$

where $K_{ij} = k(\mathbf{z}_i, \mathbf{z}_j)$. Consider evaluating $H_0 : f(\mathbf{z}_i) \equiv 0$, or equivalently $H_0 : \tau^2 = 0$. The covariance of the induced marginal model is:

$$\boldsymbol{\Sigma} = \sigma^2 \mathbf{I} + \tau^2 \mathbf{K}.$$

Identify $\boldsymbol{\alpha} = (\sigma^2, \tau^2)$. The restricted log likelihood is:

$$\ell_r(\boldsymbol{\alpha}) \propto -\frac{1}{2} \ln \det(\boldsymbol{\Sigma}) - \frac{1}{2} \ln \det(\mathbf{X}' \boldsymbol{\Sigma}^{-1} \mathbf{X}) - \frac{1}{2} \mathbf{y}' \mathbf{Q} \mathbf{y}.$$

The score equation for τ^2 is:

$$\mathcal{U}_{\tau^2}(\sigma^2, \tau^2) = \frac{\partial \ell_r}{\partial \tau^2} = -\frac{1}{2} \text{tr}(\mathbf{Q} \mathbf{K}) + \frac{1}{2} \mathbf{y}' \mathbf{Q} \mathbf{K} \mathbf{Q} \mathbf{y}.$$

Under $H_0 : \tau^2 = 0$, the error projection reduces to:

$$\mathbf{Q} = \sigma^{-2} \{ \mathbf{I} - \mathbf{X}(\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \} = \sigma^{-2} \mathbf{P}_X^\perp.$$

The score at $\tau^2 = 0$ is:

$$\mathcal{U}_{\tau^2}(\sigma^2, \tau^2 = 0) = -\frac{1}{2\sigma^2} \text{tr}(\mathbf{P}_X^\perp \mathbf{K}) + \frac{1}{2\sigma^4} \mathbf{y}' \mathbf{P}_X^\perp \mathbf{K} \mathbf{P}_X^\perp \mathbf{y}.$$

Since the trace term does not depend on $\mathbf{y}$, consider the test statistic:

$$T_K = \mathbf{y}' \mathbf{P}_X^\perp \mathbf{K} \mathbf{P}_X^\perp \mathbf{y}.$$

Let $\mathbf{L}\mathbf{L}'$ denote the Cholesky decomposition of the kernel matrix $\mathbf{K}$. Under $H_0 : \tau^2 = 0$, $\mathbf{P}_X^\perp \mathbf{y} \sim N(\mathbf{0}, \sigma^2 \mathbf{P}_X^\perp)$. Thus T_K follows a mixture of central χ_1^2 distributions:

$$T_K \sim \sum_{i=1}^n \lambda_i \chi_1^2,$$

where (λ_i) are eigenvalues of the matrix:

$$\boldsymbol{\Xi} = \mathbf{L}' \mathbb{V}(\mathbf{P}_X^\perp \mathbf{y}) \mathbf{L} = \sigma^2 \mathbf{L}' \mathbf{P}_X^\perp \mathbf{L}.$$

